

From Z to T

Roadmap

Reduced conditions for
inference about a mean

the t-test and t-distribution

Confidence intervals based
on t

Example and t-testing in R

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Part 1 of the course looked at visualizing and describing data

- ▶ Continuous(histograms, box plots, mean, median, variance, standard deviation etc.)
- ▶ Categorical (bar charts, stacked bars, frequencies/percents marginal and conditional probabilities)

Part II introduced key concepts in probability and distributions

- ▶ Probability rules (independence, addition and decomposition, multiplication, Bayes theorem)
- ▶ Continuous distribution (Normal)
- ▶ Discrete distributions (Binomial and Poisson)
- ▶ Sampling variability, central limit theorem, CI and hypothesis testing

Part III will put these together and build your toolkit for statistical testing

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In deciding what statistical test to use we will often be thinking about:

- ▶ The type of data (continuous vs categorical)
- ▶ How many groups we are comparing
- ▶ Is there an inherent relationship between the measurements (dependence)?
- ▶ Is there a theoretical distribution that is a good fit for the data?

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- ▶ Data are a SRS from a much larger population (really important)
- ▶ Observations follow a Normal distribution (some leeway)

Recap: Z testing

- ▶ We have been looking at variables that are continuous in nature
- ▶ For the last few lectures we have assumed that the population standard deviation (σ) was known to us
- ▶ We conducted the z-test and created CIs using this known σ
- ▶ Today we will generalize this framework to a more realistic setting where σ is unknown. We will use s , the sample standard deviation, as an estimate of σ

Estimating the standard error based on the sample

- ▶ Previously, we knew the standard error of the mean to be

$$\frac{\sigma}{\sqrt{n}}$$

- ▶ Now, we don't know σ , so we estimate the standard error by

$$\frac{s}{\sqrt{n}}$$

where s is the sample standard deviation.

Standard deviation vs standard error

- ▶ Variance (σ^2 for a population, s^2 for a sample): average squared deviation from the mean
- ▶ Standard deviation (σ for a population, s for a sample): the square root of the variance of all observations
 - ▶ on average, how far do individual values fall from the mean?
- ▶ Standard error (SE): the standard deviation of a *statistic* computed from the data. For the sample mean it is $s/\sqrt{n}$
 - ▶ it describes the spread of $\bar{x}$ across all possible samples
 - ▶ on average, how far does our sample mean fall from the population mean?

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Recall the z-test!

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$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

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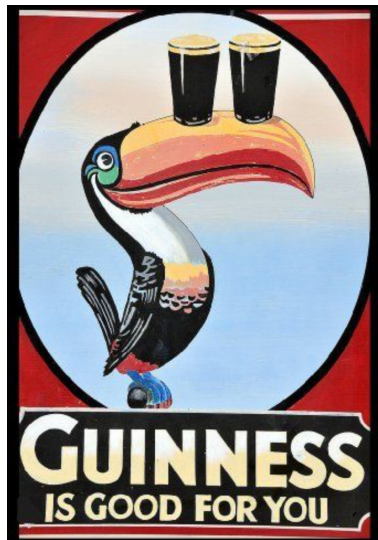
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Meet the t-test!

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Thanks to William Gosset (published anonymously under the pen name “Student”) and the Guinness company’s strategy of hiring statisticians, if we are interested in comparing mean values of a variable to a hypothesized null we can use a t-test.

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

- ▶ What is the difference between z and t?
- ▶ The t-test is more variable than the z-test statistic because we have to substitute s for σ . Because s is a statistic, it varies across samples.
- ▶ Because of this substitution, the t-test will not follow a $\text{Normal}(0, 1)$ distribution. It is *more* variable than the standard Normal. Thus, we need a distribution that is like the standard Normal but a little bit wider.

What do we know about our estimate of the mean and its variability as the sample size grows?

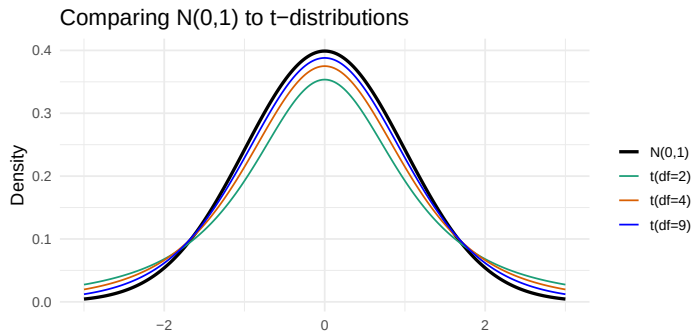
- ▶ The standard error $s/\sqrt{n}$ shrinks, so $\bar{x}$ lands closer to μ
- ▶ s itself becomes a more reliable estimate of σ , so the extra uncertainty from substituting s for σ gets smaller
- ▶ With a large enough n , using s is almost as good as knowing σ

Introducing the t distribution

- ▶ Like the standard Normal distribution, but wider.
- ▶ Its width depends on n , the sample size, which determines the **degrees of freedom**

This is because as n increases, our estimate s gets better and better, and approaches σ . Thus, as n increases the t-distribution approaches a Normal(0, 1) distribution.

Introducing the t distribution



The t statistic and degrees of freedom

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$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

The one-sample t statistic has a t distribution with $n - 1$ **degrees of freedom**

You calculate the t statistic using $\bar{x}$ and s estimated from your sample, and n which is also a property of your sample, and μ_0 from the null hypothesis. Then compute the probability of observing a value of t or more extreme.

Assumptions of the t-test

- ▶ The dependent variable must be continuous (interval/ratio).
- ▶ The observations are independent of one another.
- ▶ Data come from a random sample of the underlying population.
- ▶ The dependent variable should be approximately normally distributed.
- ▶ The dependent variable should not contain any outliers.

Guess the R functions

```
pt(q = , df = , lower.tail = )  
qt(p = , df = , lower.tail = )
```

Which one would we use to calculate the p-value for a hypothesis test after we calculated the t-test statistic? pt or qt?

Guess the R functions: answer

`pt()` takes a **quantile** (your observed t statistic) and returns a **probability**. That is what a p-value is, so we use `pt`.

`qt()` goes the other way. It takes a probability and returns the quantile, which is what we need for the critical value t^* in a confidence interval.

```
pt(q = 2.5, df = 17, lower.tail = FALSE) # a p-value
```

```
## [1] 0.0114739
```

```
qt(p = 0.975, df = 17) # a critical value
```

```
## [1] 2.109816
```

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Calculating a confidence interval for the t-test

Draw an SRS of size n from a large population having unknown mean μ and unknown standard deviation σ . A level C **confidence interval for μ** is:

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

where t^* is the critical value for the $t(n-1)$ density curve with area C between $-t^*$ and t^* .

Supposing we had $n = 100$, what is t^* for a 95% confidence interval?

```
qt(p = 0.975, df = 99)
```

```
## [1] 1.984217
```

Compare that to $z^* = 1.96$. With $n = 100$ the t critical value is only a little wider.

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Example: Testosterone and obesity in adolescent males (pg 422 B&M Ed 4)

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Here are the data for $n = 25$ adolescent males between the ages of 14 and 20:

```
library(dplyr)
testosterone <- c(0.30, 0.24, 0.19, 0.17, 0.18, 0.23, 0.24, 0.06, 0.15,
                  0.17, 0.18, 0.17, 0.15, 0.12, 0.25, 0.25, 0.25, 0.32,
                  0.35, 0.37, 0.39, 0.46, 0.49, 0.42, 0.36)
dat_test <- data.frame(testosterone)
```


Example: Testosterone and obesity in adolescent males (pg 422 B&M Ed 4)

Use R to calculate a 95% confidence interval for testosterone. We can do this using `summarize`

```
dat_test %>% summarize(sample_mean = mean(testosterone),  
                        sample_sd = sd(testosterone),  
                        sample_size = length(testosterone),  
                        sample_se = sample_sd/sqrt(sample_size))
```

```
##   sample_mean sample_sd sample_size sample_se  
## 1      0.2584 0.1115303         25 0.02230605
```

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We still need the t^* value:

```
t_star <- qt(p = 0.975, df = 24)
t_star
```

```
## [1] 2.063899
```

Example: Testosterone and obesity in adolescent males (pg 422 B&M Ed 4)

Expand the previous code chunk to calculate the margin of error (which uses the critical t^* value), and then calculate the lower and upper CI

```
dat_test %>% summarize(sample_mean = mean(testosterone),
                        sample_sd = sd(testosterone),
                        sample_size = length(testosterone),
                        sample_se = sample_sd/sqrt(sample_size),
                        margin_of_error = sample_se*t_star,
                        lower_CI = sample_mean - margin_of_error,
                        upper_CI = sample_mean + margin_of_error)

##      sample_mean sample_sd sample_size  sample_se margin_of_error
## 1          0.2584 0.1115303          25 0.02230605      0.04603743
##      lower_CI  upper_CI
## 1 0.2123626 0.3044374
```

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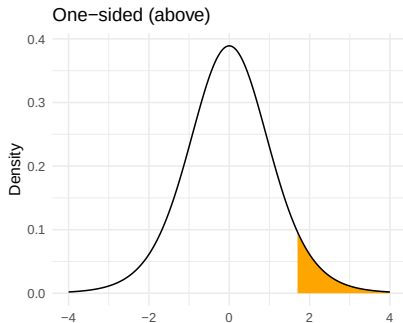
Draw an SRS of size n from a large population having unknown mean μ and unknown standard deviation σ . To test the hypothesis $H_0 : \mu = \mu_0$, calculate the t statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Hypothesis testing with unknown σ using the t-test

In terms of a variable T having the $t(n-1)$ distribution, the p-value for a test of H_0 against

$$H_a: \mu > \mu_0 \text{ is } P(T \geq t)$$



Hypothesis testing with unknown σ using the t-test

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$$H_a: \mu < \mu_0 \text{ is } P(T \leq t)$$



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$$H_a: \mu \neq \mu_0 \text{ is } 2 \times P(T \geq |t|)$$



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Example of a t-test (pg 426 B&M Ed 4)

Here are 18 measures of pulse wave velocity (PWV) from a sample of children diagnosed with progeria, a genetic disorder that produces rapid aging.

```
pwv <- c(18.8, 17.6, 17.5, 16.0, 14.8, 14.1, 13.7, 13.1, 12.9,  
         12.9, 12.4, 10.1, 9.3, 9.1, 8.3, 8.3, 7.9, 7.2)
```

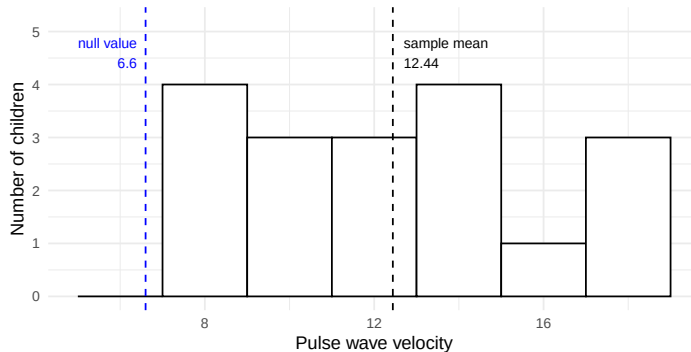
```
pwv_dat <- data.frame(pwv)
```

Example of a t-test (pg 426 B&M Ed 4)

pwv measures greater than 6.6 are considered abnormally high. We would like to test the hypothesis that the mean for this subset of children is abnormally high.

That is: $H_0 : \mu = 6.6$ and $H_a : \mu > 6.6$

Look at the data and see if there is evidence against the null hypothesis



Example of a one-sided t-test (pg 426 B&M Ed 4)

```
pwv_dat %>%  
  summarize(sample_mean = mean(pwv),  
            sample_sd = sd(pwv),  
            sample_size = length(pwv),  
            sample_se = sample_sd/sqrt(sample_size),  
            t_test = (sample_mean - 6.6)/sample_se,  
            p_value = 1 - pt(t_test, df = sample_size - 1))
```

```
##      sample_mean sample_sd sample_size sample_se  t_test  
## 1      12.44444  3.637747          18 0.8574252 6.816273  
##      p_value  
## 1 1.501248e-06
```

There's a function for that...

Rather than doing the test using `summarize`, we could have R do it for us using `t.test`:

```
t.test(x = pwv_dat %>% pull(pwv), alternative = "greater", mu = 6.6)
```

```
##  
## One Sample t-test  
##  
## data:  pwv_dat %>% pull(pwv)  
## t = 6.8163, df = 17, p-value = 1.501e-06  
## alternative hypothesis: true mean is greater than 6.6  
## 95 percent confidence interval:  
##  10.95286      Inf  
## sample estimates:  
## mean of x  
##  12.44444
```

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- ▶ A confidence interval or hypothesis test is called **robust** if the confidence level or P-value does not change very much when the conditions for use of the procedure are violated.
- ▶ In particular, how robust are the procedures against non-Normality?
- ▶ The t procedures are quite robust against non-Normality of the population except when outliers or strong skewness are present.
- ▶ The t procedures are not robust against outliers unless the sample size is sufficiently large.

Checking assumptions

- ▶ Always plot your data first:
 - ▶ Are there any outliers?
 - ▶ Is the distribution of the data skewed?

Guidelines for using the t procedures

- ▶ The SRS condition is more important than the Normality condition
- ▶ If $n < 15$: Use t procedures if the data appear close to Normal (at least roughly symmetric, single peak, no outliers). If the data are skewed or there are outliers, don't use t .
- ▶ Moderate sample size > 15 : The t procedures can be used except in the presence of outliers or strong skewness
- ▶ Large sample size, roughly $n \geq 40$: The t procedures can be used even for strongly skewed distributions when the sample is large, roughly $n \geq 40$

Example 17.5: Can we use t ? (1 of 4)

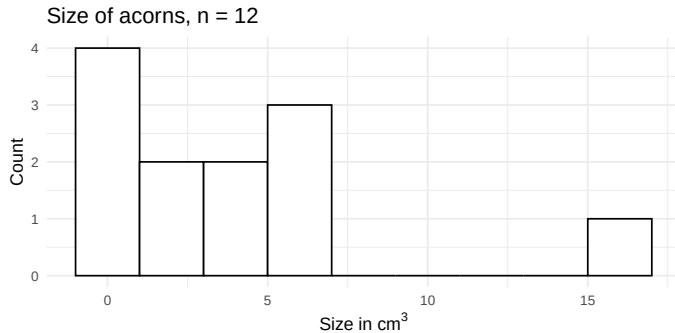
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Example 17.5: Can we use t ? (2 of 4)

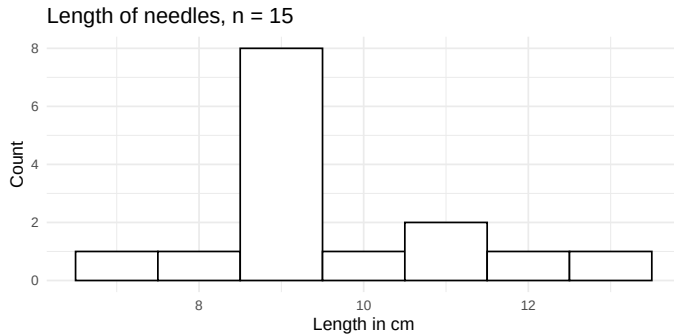
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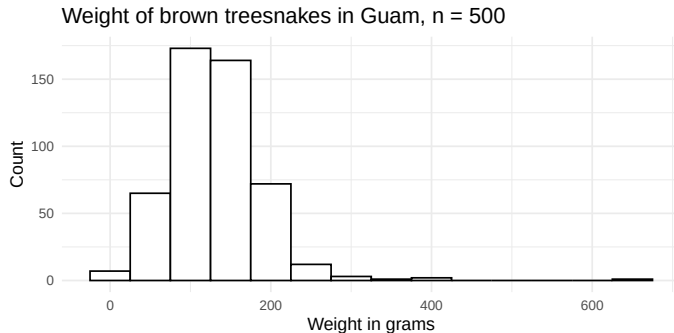
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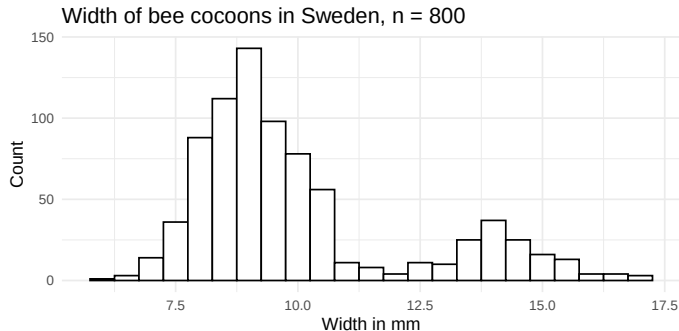
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Example 17.5: Can we use t ? (3 of 4)



Example 17.5: Can we use t ? (4 of 4)



Can we use t ? Answers

- ▶ Acorns, $n = 12$: no. Small sample, right skewed, and a clear outlier at 16.5.
- ▶ Needles, $n = 15$: yes. Roughly symmetric, single peak, no outliers.
- ▶ Treesnakes, $n = 500$: yes. Skewed with a high outlier, but n is far above 40.
- ▶ Bee cocoons, $n = 800$: yes for t , but the two peaks suggest two subgroups, so a single mean may not be the quantity you actually care about.

Sample size rescues skewness. It does not rescue a bad question.

- ▶ When σ is unknown we plug in s , and the extra uncertainty means our statistic follows a $t(n-1)$ distribution rather than $N(0,1)$
- ▶ $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ and $CI = \bar{x} \pm t^* \frac{s}{\sqrt{n}}$
- ▶ In R: `pt()` for p-values, `qt()` for critical values, or `t.test()` for both
- ▶ Check the conditions by plotting first. SRS matters most, then outliers and skewness relative to n

Parting Humor

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